Introduction to Telecommunications (EN1052)

Problem Session
February 2021

Problem 1

Signal multiplication at higher frequencies can be accomplished by the square-law modulator shown in Figure Q1, where $x(t)$ is the *real* input of the modulator and $y(t)$ is the output. The block ND represents a *nonlinear element*, whose input $v_{\text{in}}(t)$ and output $v_{\text{out}}(t)$ are related by

$$v_{\text{out}}(t) = v_{\text{in}}(t) + 0.25v_{\text{in}}^2(t).$$

The system $\mathcal{H}$ is a linear time-invariant filter. It is given that the Fourier transforms of all the associated signals exist.

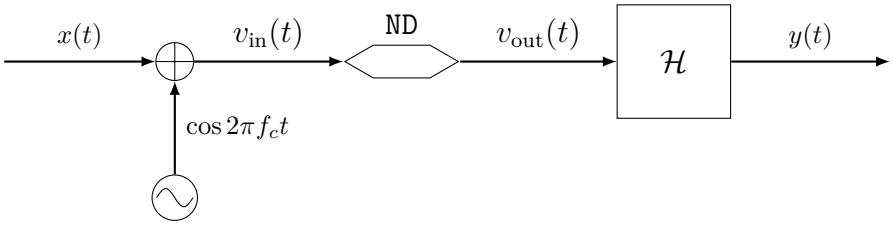


Figure 1: Square-law modulator.

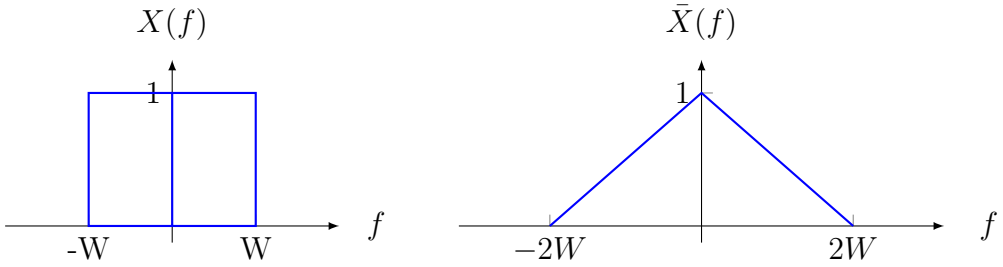


Figure 2: Spectrums of $x(t)$ and $x^2(t)$.

1. As shown in Figire 1, we have $v_{\text{in}}(t) = x(t) + \cos 2\pi f_c t$. Determine the output $v_{\text{out}}(t)$ of ND.

2. Let the spectrums of $x(t)$ and $x^2(t)$ are denoted by $X(f)$ and $\bar{X}(f)$, respectively, see Figure 2. Determine the spectrum $V_{\text{out}}(f)$ of $v_{\text{out}}(t)$.

   [*Hint: See the* ***Appendix*** *on the last page for a short list of Fourier transform pairs.*]

3. Sketch the spectrum $V_{\text{out}}(f)$ for

   (a) $f_c = 2W$ Hz,

   (b) $f_c = 3W$ Hz,

(c) $f_c = 4W$ Hz.

4. By considering the answer to part (3), explain how the square-law modulator can be used to realize a *standard amplitude modulation* scheme by clearly identifying the role of $\mathcal{H}$. You may use sketches of the spectrum $V_{\text{out}}(f)$ and the frequency response of $\mathcal{H}$ whenever necessary.

6 points

Problem 2

Suppose 6 real message signals $x_1(t), x_2(t), \ldots, x_6(t)$ of bandwidth 6000 Hz, 1400 Hz, 1300 Hz, 1200 Hz, 600 Hz, and 400 Hz, respectively, are to be multiplexed in time. To do this, an n-channel multiplexer (MUX) with input frequency f_s Hz is used. The parameters n and f_s are to be determined according to the application requirements. Each channel (`ch`) of the MUX must be fed with pulse-amplitude modulation (PAM) pulses at a rate f_s Hz. For example, the functionality of a 8-channel MUX with $f_s = 1000$ Hz is depicted in Figure 3. Note that the PAM pulses generated by each channel at a rate 1000 Hz are combined to create a pulse rate of $8 \times 1000 = 8000$ Hz, see $m(t)$ in Figure 3

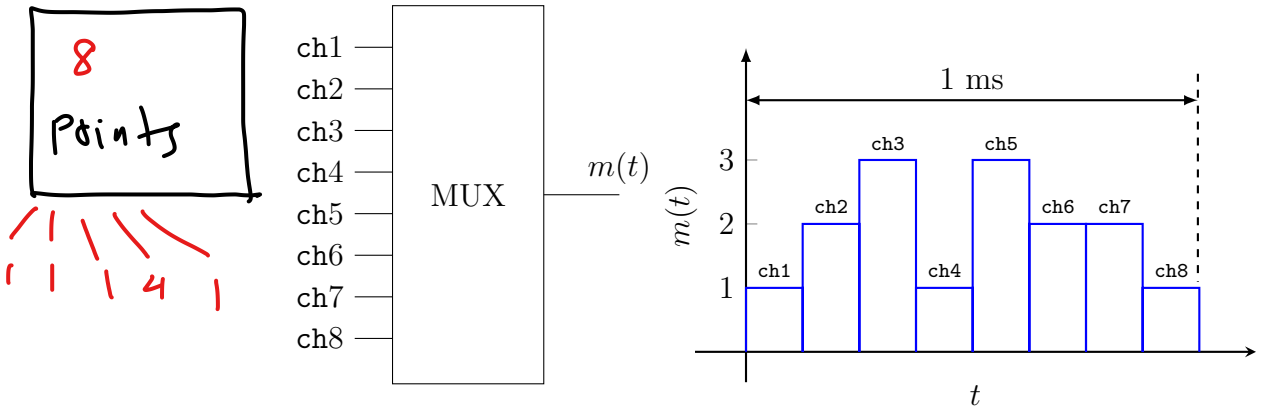


Figure 3: Multiplexer with $n = 8$ and $f_s = 1000$ Hz.

1. Suppose all the signals $x_1(t)$ to $x_6(t)$ are to be sampled at the *same sampling rate*. Determine the *minimum* required sampling rate f_N if the signals are to be reconstructed at their respective receivers with *no aliasing*.

2. Given that the sampling rate f_N [see part (1)] of the signals equals to their corresponding PAM pulse rate. Moreover, the PAM pulses related to each signal are to be multiplexed by using a 6-channel MUX.

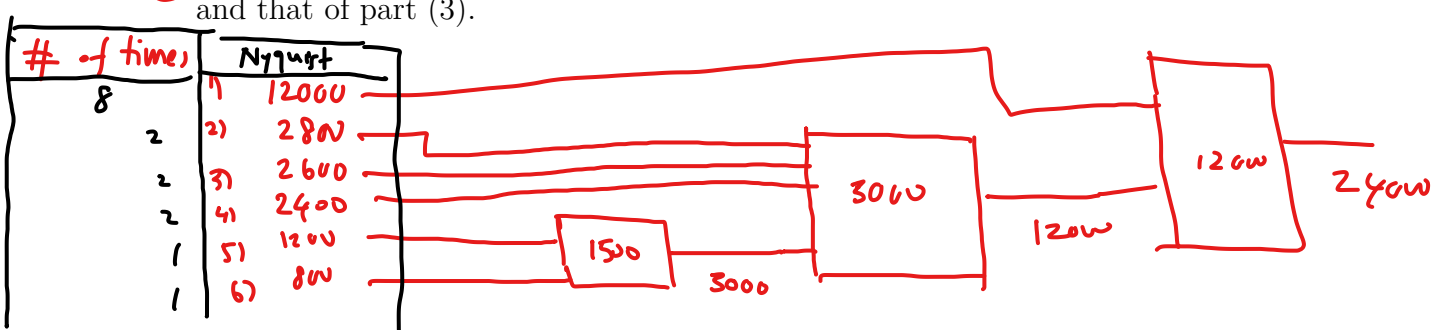
   (a) Compute the input frequency f_s of the MUX.

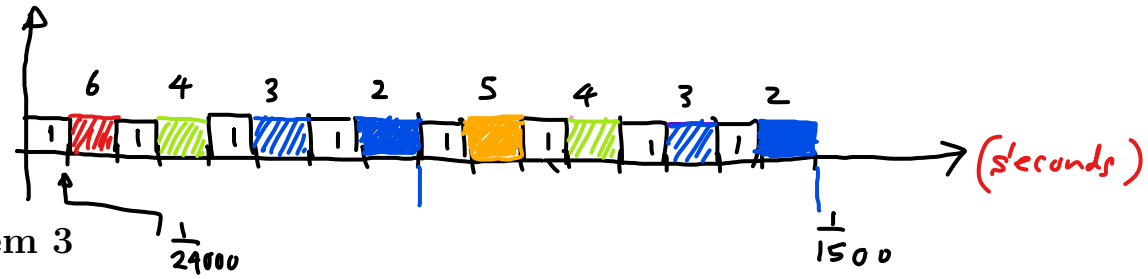
   (b) Sketch the MUX configuration by clearly identifying the inputs.

   (c) Determine the pulse rate at the output of the MUX.

3. Suppose you are give MUXes of 3 types; type 1 is with $n = 2$, $f_s = 1500$ Hz, type 2 is with $n = 4$, $f_s = 3000$ Hz, and type 3 is with $n = 2$, $f_s = 12000$ Hz. Sketch the MUX configuration to multiplex the signals $x_1(t)$ to $x_6(t)$ with *no aliasing* at their receivers. You have to clearly identify the sampling rates of the signals, the inputs of each MUX, and their interconnections.

4. Compare and contrast the pulse rate at the output of MUX configuration in part (2) and that of part (3).





Problem 3

Frequency division multiplexing (FDM) is a key operation in any analog or digital communication system for bundling baseband message signals from different entities. An example of a typical FDM transmitter schematic, where the *real* message signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ individually modulate the *subcarriers* f_1, f_2, and f_3, respectively is shown in Figure 4. Moreover, the Double-sideband suppressed-carrier (DSB-SC) modulation is employed. The resulting signals $y_1(t)$, $y_2(t)$, and $y_3(t)$, at the end of each subcarrier modulation stage are then summed to produce $x_b(t)$. The final carrier modulation at a frequency f_c is employed to yield $x_c(t)$, where $f_c \gg f_1, f_2, f_3$.

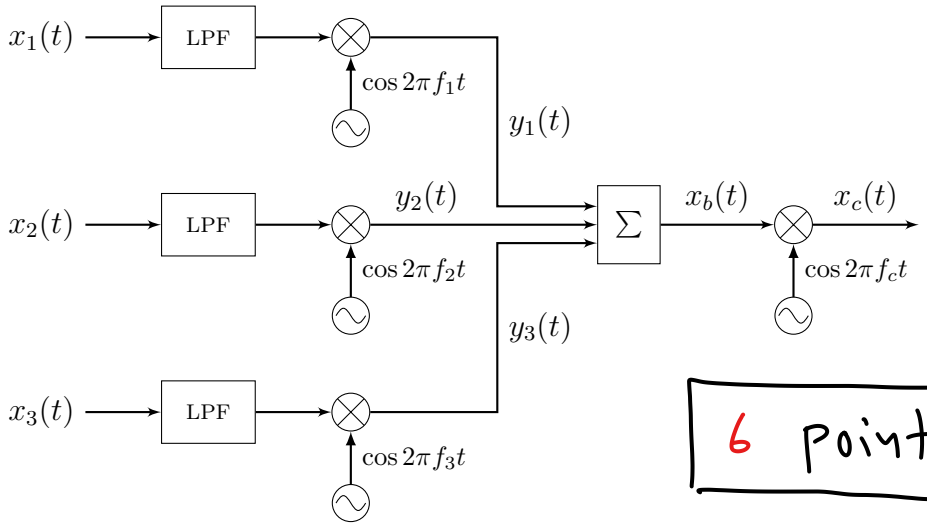


Figure 4: Typical FDM transmitter.

1. Explain why the FDM transmitter should employ low pass filter (LPF) blocks as shown in Figure 4.

2. Suppose you are given n real message signals $x_1(t), x_2(t), \ldots, x_n(t)$. The signals are to be multiplexed by using an FDM transmitter of the form Figure 4. In this respect, a proportional bandwidth DSB-SC subcarrier modulation is employed, where the ith subcarrier frequency f_i and the bandwidth W_i of the ith message signal are related by $f_i = \alpha W_i$ for some positive constant α. It is given that $W_i \neq W_j$ for all i, j, $i \neq j$. Without loss of generality, it is assumed that $W_1 < W_2 < \cdots < W_n$. In order to ensure *no overlapping* of the associated spectrums of x_is in the transmitted signal $x_c(t)$, show that α should satisfy

$$\alpha \geq \frac{1+\beta}{1-\beta},$$

where β is the maximum of ratios $W_1/W_2, W_2/W_3, \ldots, W_{n-1}/W_n$.

[*Hint: Draw the spectrum of $x_b(t)$ to identify possible overlapping regions and check conditions to avoid them.*]

3. If $n = 3$, $W_1 = 100$ Hz, $W_2 = 200$ Hz, and $W_3 = 250$ Hz, compute an applicable value for α and the associated subcarrier frequencies f_1, f_2, and f_3.

$\alpha(\beta) + \beta \leq (\alpha - 1)$

$(1+\beta) \leq \alpha(1-\beta)$

$\alpha \geq \frac{(1+\beta)}{(1-\beta)}$

$\frac{w_i}{w_{i+1}} \leq \frac{(\alpha-1)}{(\alpha+1)} \quad \forall i \quad \beta \leq \left(\frac{\alpha-1}{\alpha+1}\right)$

$\Rightarrow \left(\max_i \frac{w_i}{w_{i+1}}\right) \leq \left(\frac{\alpha-1}{\alpha+1}\right)$

Problem 4

Not considered

1. An alphanumeric message has 300-characters and it should be transmitted in 2 s. Each character is to be encoded by using 7-bit ASCII coding, followed by a bit (i.e., the eighth bit) for error detection *per character*.

   (a) If a multilevel pulse-amplitude modulation (PAM) waveform with $M = 8$ levels is used, calculate the effective transmitted bit rate and the symbol rate.

   (b) Repeat part (i) for pulse code modulation (PCM) waveforms.

2. The information in an analog waveform, whose maximum frequency 4000 Hz, is to be transmitted. The maximum quantization distortion $e_{\max}$ must not exceed $\pm 1\%$ of the peak-to-peak voltage V_{pp} of the analog signal, that is $|e_{\max}| \leq 0.01 V_{\text{pp}}$.

   (a) What is the minimum number of bits per sample that should be used in this transmission system?

   [*Hint: You may use the relation* $|e_{\max}| = V_{\text{pp}}/(2L)$, *where* L *is the number of quantization levels.*]

   (b) What is the minimum required sampling rate?

   (c) Compute the resulting bit rate.

Appendix

Some useful Fourier transform pairs are given in the following table.

	$g(t)$	$G(f) = \mathcal{F}[g(t)]$
1.	1	$\delta(f)$
2.	$e^{j2\pi f_0 t}$	$\delta(f - f_0)$
3.	$\cos 2\pi f_0 t$	$\frac{1}{2}\left[\delta(f - f_0) + \delta(f + f_0)\right]$
4.	$g(t)\cos 2\pi f_0 t$	$\frac{1}{2}\left[G(f - f_0) + G(f + f_0)\right]$